

Codingan Sequence Diagram PlantUML

Spesifikasi UML 16 Use Case KuroFin - Standar RPL Pertemuan 04 (3-Tier ECB Architecture)

UC-01: Melihat Dashboard & Ringkasan Saldo

Menampilkan saldo total, akumulasi pemasukan, pengeluaran, dan status ringkasan saldo.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "DashboardUI" as UI
control "DashboardController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-01: Melihat Dashboard & Ringkasan Saldo

User -> UI: bukaAplikasiKuroFin()
activate UI
UI -> Controller: loadDashboardData()
activate Controller

Controller -> DB: fetchAllTransactions()
activate DB

alt Data Transaksi Tersedia
    DB --> Controller: dataTransaksiList
    Controller -> Controller: calculateTotalBalance(pemasukan - pengeluaran)
    Controller --> UI: displaySummary(saldoTotal, totalPemasukan, totalPengeluaran)
    UI --> User: tampilkanRingkasanSaldo()
else Basis Data Kosong
    DB --> Controller: dataEmpty
    Controller --> UI: displaySummary(saldo: 0)
    UI --> User: tampilkanSaldoNol()
else Gagal Membaca File Database
    DB --> Controller: errorReadAccess
    deactivate DB
    Controller --> UI: showErrorAlert("Gagal Membaca Basis Data")
    deactivate Controller
    UI --> User: tampilkanPesanErrorAkses()
    deactivate UI
end
@enduml
```

UC-02: Mencatat Transaksi Baru

Pencatatan transaksi yang terintegrasi dengan UC-06 (Validasi), UC-07 (Update Saldo), dan UC-11 (Budget Alert).

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "TransactionFormUI" as UI
control "TransactionController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-02: Mencatat Transaksi Baru

User -> UI: bukaFormInput()
activate UI
UI --> User: tampilkanFormInput()

User -> UI: inputDataAndKlikSimpan(tanggal, jenis, kategori, nominal, ket)
UI -> Controller: simpanTransaksi(data)
activate Controller

note over Controller: UC-06: Validasi Input & Kategori
Controller -> Controller: validasiInputAndKategori(data)

alt Data Valid (Nominal > 0 & Kategori Ada)
    Controller -> DB: insertTransaction(data)
    activate DB
    DB --> Controller: statusSukses
    deactivate DB

    note over Controller: UC-07: Pembaruan Saldo Otomatis
    Controller -> DB: recalculateTotalBalance()
    activate DB
    DB --> Controller: balanceUpdated
    deactivate DB

    opt Jenis Transaksi = Pengeluaran
        note over Controller: UC-11: Evaluation Budget Alert
        Controller -> DB: checkBudgetLimit()
        activate DB
        DB --> Controller: budgetStatus
        deactivate DB
        opt Pengeluaran >= 80% Budget Limit
            Controller -> UI: showBudgetAlertNotification()
        end
    end

    Controller --> UI: pesanSukses("Transaksi Berhasil Disimpan")
    UI --> User: tampilkanNotifikasiSuksesAndResetForm()

else Data Tidak Valid (Nominal <= 0 / Error)
    Controller --> UI: pesanError("Input Tidak Valid")
    deactivate Controller
    UI --> User: tampilkanPesanPeringatanError()
    deactivate UI
end
@enduml
```

UC-03: Melihat Riwayat Transaksi

Menampilkan seluruh riwayat transaksi diurutkan berdasarkan tanggal terbaru.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "TransactionHistoryUI" as UI
control "TransactionController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-03: Melihat Riwayat Transaksi

User -> UI: bukaMenuRiwayatTransaksi()
activate UI
UI -> Controller: getTransactionHistory()
activate Controller

Controller -> DB: selectTransactionsSortedByDateDESC()
activate DB

alt Terdapat Rekaman Transaksi
    DB --> Controller: transactionList
    Controller --> UI: renderTransactionTable(transactionList)
    UI --> User: tampilkanTabelRiwayat()
else Belum Ada Transaksi
    DB --> Controller: emptyList
    deactivate DB
    Controller --> UI: renderEmptyTableMessage("Belum Ada Transaksi")
    deactivate Controller
    UI --> User: tampilkanKeteranganKosong()
    deactivate UI
end
@enduml
```

UC-04: Mengubah Data Transaksi

Penyuntingan transaksi yang memperbarui database dan saldo.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "EditTransactionUI" as UI
control "TransactionController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-04: Mengubah Data Transaksi

User -> UI: pilihTransaksiAndKlikEdit(id)
activate UI
UI -> Controller: getTransactionById(id)
activate Controller
Controller -> DB: selectById(id)
activate DB
DB --> Controller: transactionDetail
deactivate DB
Controller --> UI: populateForm(transactionDetail)
deactivate Controller

User -> UI: ubahDataAndKlikSimpanPerubahan(newData)
UI -> Controller: updateTransaksi(id, newData)
activate Controller

note over Controller: UC-06: Validasi Input
Controller -> Controller: validasiInputAndKategori(newData)

alt Data Valid
    Controller -> DB: updateTransactionRecord(id, newData)
    activate DB
    DB --> Controller: statusUpdated
    deactivate DB

    note over Controller: UC-07: Pembaruan Saldo Otomatis
    Controller -> DB: recalculateTotalBalance()
    activate DB
    DB --> Controller: balanceUpdated
    deactivate DB

    Controller --> UI: pesanSukses("Data Berhasil Diperbarui")
    UI --> User: tampilkanNotifikasiSukses()
else Data Tidak Valid
    Controller --> UI: pesanError("Nilai Tidak Valid, Data Lama Dipertahankan")
    deactivate Controller
    UI --> User: tampilkanErrorValidasi()
    deactivate UI
end
@enduml
```

UC-05: Menghapus Data Transaksi

Penghapusan catatan transaksi berbasis dialog konfirmasi.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "TransactionHistoryUI" as UI
control "TransactionController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-05: Menghapus Data Transaksi

User -> UI: pilihTransaksiAndKlikHapus(id)
activate UI
UI --> User: tampilkanDialogKonfirmasi("Hapus Data Ini?")

alt Pengguna Menekan "Ya"
    User -> UI: konfirmasiHapus("Ya")
    UI -> Controller: deleteTransaction(id)
    activate Controller
    Controller -> DB: deleteTransactionRecord(id)
    activate DB
    DB --> Controller: statusDeleted
    deactivate DB

    note over Controller: UC-07: Pembaruan Saldo Otomatis
    Controller -> DB: recalculateTotalBalance()
    activate DB
    DB --> Controller: balanceUpdated
    deactivate DB

    Controller --> UI: pesanSukses("Transaksi Berhasil Dihapus")
    UI --> User: perbaruiTabelAndTampilkanPesanSukses()
else Pengguna Menekan "Tidak"
    User -> UI: konfirmasiHapus("Tidak")
    UI --> User: tutupDialogAndBatalHapus()
    deactivate UI
end
@enduml
```

UC-06: Validasi Input & Kategori (Included UC)

Proses internal pemeriksaan tipe angka, nominal, dan ketersediaan kategori.

```
@startuml
autonumber
participant "CallerController" as Caller
control "ValidationControl" as Val
entity "Database SQLite" as DB

title Sequence Diagram UC-06: Validasi Input & Kategori (Included)

Caller -> Val: validateTransactionInput(data)
activate Val

Val -> Val: checkNominalPositif(data.nominal)
alt Nominal <= 0 Atau Bukan Angka
    Val --> Caller: returnStatus(INVALID_NOMINAL)
else Nominal Valid (> 0)
    Val -> DB: verifyCategoryExists(data.kategoriId)
    activate DB
    alt Kategori Terdaftar
        DB --> Val: categoryExists
        Val --> Caller: returnStatus(VALID_SUCCESS)
    else Kategori Tidak Terdaftar
        DB --> Val: categoryNotFound
        deactivate DB
        Val --> Caller: returnStatus(INVALID_CATEGORY)
        deactivate Val
    end
end
end
@enduml
```

UC-07: Pembaruan Saldo Otomatis (Included UC)

Penghitungan selisih total pemasukan dan pengeluaran secara real-time.

```
@startuml
autonumber
participant "TransactionController" as Trans
control "BalanceController" as Bal
entity "Database SQLite" as DB
boundary "DashboardUI" as UI

title Sequence Diagram UC-07: Pembaruan Saldo Otomatis (Included)

Trans -> Bal: triggerBalanceUpdate()
activate Bal

Bal -> DB: sumTotalPemasukan()
activate DB
DB --> Bal: totalPemasukan
Bal -> DB: sumTotalPengeluaran()
DB --> Bal: totalPengeluaran
deactivate DB

Bal -> Bal: calculateBalance(totalPemasukan - totalPengeluaran)
Bal -> UI: updateBalanceLabel(totalSaldo)
activate UI
UI --> Bal: labelUpdated
deactivate UI
Bal --> Trans: balanceUpdateComplete
deactivate Bal
@enduml
```

UC-08: Mengelola Kategori Transaksi

Operasi Tambah, Ubah, dan Hapus kategori transaksi.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "CategoryFormUI" as UI
control "CategoryController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-08: Mengelola Kategori Transaksi

User -> UI: bukaHalamanKategori()
activate UI

alt Aksi: Tambah Kategori
    User -> UI: inputNamaKategoriAndKlikTambah(nama)
    UI -> Controller: addCategory(nama)
    activate Controller
    Controller -> DB: checkDuplicateCategoryName(nama)
    activate DB
    alt Nama Belum Ada (Valid)
        DB --> Controller: nameAvailable
        Controller -> DB: insertCategory(nama)
        DB --> Controller: statusInserted
        Controller --> UI: pesanSukses("Kategori Berhasil Ditambahkan")
    else Nama Duplikat
        DB --> Controller: nameExists
        Controller --> UI: pesanError("Nama Kategori Sudah Ada")
    end
end

else Aksi: Ubah Kategori
    User -> UI: pilihKategoriAndUbahNama(id, namaBaru)
    UI -> Controller: updateCategory(id, namaBaru)
    Controller -> DB: updateCategoryRecord(id, namaBaru)
    DB --> Controller: statusUpdated
    Controller --> UI: pesanSukses("Kategori Diperbarui")
end

else Aksi: Hapus Kategori
    User -> UI: pilihKategoriAndKlikHapus(id)
    UI -> Controller: deleteCategory(id)
    Controller -> DB: checkCategoryTransactionRelation(id)
    alt Kategori Bebas (Tidak Terikat Transaksi)
        DB --> Controller: noRelation
        Controller -> DB: deleteCategoryRecord(id)
        DB --> Controller: statusDeleted
        Controller --> UI: pesanSukses("Kategori Dihapus")
    else Kategori Masih Digunakan Transaksi
        DB --> Controller: relationExists
        deactivate DB
        Controller --> UI: pesanWarning("Kategori Masih Digunakan Transaksi")
        deactivate Controller
    end
end

end
UI --> User: tampilkanDaftarKategoriTerbarui()
deactivate UI
@endstartuml
```

UC-09: Mengatur Batas Pengeluaran

Penetapan target nominal maksimal pengeluaran per periode.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "BudgetUI" as UI
control "BudgetController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-09: Mengatur Batas Pengeluaran

User -> UI: bukaHalamanPengaturanBatas()
activate UI
User -> UI: pilihTargetPeriodeAndInputNominal(periode, nominal)
UI -> Controller: saveBudgetLimit(periode, nominal)
activate Controller

alt Nominal > 0 (Valid)
    Controller -> DB: insertOrUpdateBudgetRecord(periode, nominal)
    activate DB
    DB --> Controller: statusSaved
    deactivate DB
    Controller --> UI: pesanSukses("Batas Anggaran Tersimpan")
    UI --> User: tampilkanNotifikasiBerhasil()
else Nominal <= 0 (Tidak Valid)
    Controller --> UI: pesanPeringatan("Nominal Harus Lebih Besar dari 0")
    deactivate Controller
    UI --> User: tampilkanPeringatanInput()
    deactivate UI
end

end
@enduml
```

UC-10: Memperbarui atau Menghapus Batas

Pembaruan nominal/periode atau penghapusan aturan anggaran.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "BudgetUI" as UI
control "BudgetController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-10: Memperbarui atau Menghapus Batas

User -> UI: pilihAturanBatas(budgetId)
activate UI

alt Aksi: Memperbarui Batas
    User -> UI: ubahNominalAndKlikPerbarui(budgetId, newNominal)
    UI -> Controller: updateBudgetLimit(budgetId, newNominal)
    activate Controller
    Controller -> DB: updateBudgetRecord(budgetId, newNominal)
    activate DB
    DB --> Controller: statusUpdated
    deactivate DB
    Controller --> UI: pesanSukses("Aturan Batas Diperbarui")
else Aksi: Menghapus Batas
    User -> UI: klikHapusBatas(budgetId)
    UI --> User: tampilkanKonfirmasi("Hapus Aturan Ini?")
    alt Konfirmasi Disetujui ("Ya")
        User -> UI: konfirmasiHapus("Ya")
        UI -> Controller: deleteBudgetLimit(budgetId)
        Controller -> DB: deleteBudgetRecord(budgetId)
        activate DB
        DB --> Controller: statusDeleted
        deactivate DB
        Controller --> UI: pesanSukses("Aturan Batas Dihapus")
    else Konfirmasi Ditolak ("Tidak")
        User -> UI: konfirmasiHapus("Tidak")
        UI --> User: batalHapus()
        deactivate Controller
        deactivate UI
    end
end
end
@enduml
```

UC-11: Menampilkan Notifikasi Budget Alert (Extended UC)

Pemicuan pop-up peringatan visual saat pengeluaran melampaui ambang batas.

```
@startuml
autonumber
participant "TransactionController" as Trans
control "BudgetAlertController" as Alert
entity "Database SQLite" as DB
boundary "TransactionFormUI" as UI
actor "Pengguna (User)" as User

title Sequence Diagram UC-11: Menampilkan Notifikasi Budget Alert (Extended)

Trans -> Alert: evaluateBudgetStatus(currentExpense)
activate Alert

Alert -> DB: getActiveBudgetLimit()
activate DB
DB --> Alert: budgetLimitData
Alert -> DB: calculateTotalExpenseForPeriod()
DB --> Alert: totalAccumulatedExpense
deactivate DB

Alert -> Alert: calculatePercentage(totalExpense / budgetLimit)

alt Akumulasi Pengeluaran >= 80% AND <= 100%
    Alert -> UI: showYellowAlert("Mendekati Batas Anggaran!")
    activate UI
    UI --> User: tampilkanPopUpKuningMendekatiBatas()
    User -> UI: tekanOK()
    UI --> Alert: dialogClosed
    deactivate UI
else Akumulasi Pengeluaran > 100%
    Alert -> UI: showRedAlert("Overbudget! Pengeluaran Melampaui Batas")
    activate UI
    UI --> User: tampilkanPopUpMerahOverbudget()
    User -> UI: tekanOK()
    UI --> Alert: dialogClosed
    deactivate UI
end
Alert --> Trans: evaluationComplete
deactivate Alert
@enduml
```

UC-12: Melihat Laporan Keuangan

Penyajian rekapitulasi data keuangan berdasarkan periode.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "ReportUI" as UI
control "ReportController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-12: Melihat Laporan Keuangan

User -> UI: bukaHalamanLaporan()
activate UI
User -> UI: pilihFilterPeriodeAndKlikTampilkan(startDate, endDate)
UI -> Controller: generateFinancialReport(startDate, endDate)
activate Controller

Controller -> DB: queryAgregasiKeuangan(startDate, endDate)
activate DB

alt Data Transaksi Pada Periode Ada
    DB --> Controller: reportDataAgregat
    Controller --> UI: renderReportTables(pemasukan, pengeluaran, saldo)
    UI --> User: tampilkanTabelRekapitulasiLaporan()
else Data Transaksi Kosong
    DB --> Controller: noDataFound
    deactivate DB
    Controller --> UI: renderEmptyReportMessage("Tidak Ada Transaksi pada Periode Ini")
    deactivate Controller
    UI --> User: tampilkanPesanLaporanKosong()
    deactivate UI
end
@enduml
```

UC-13: Melihat Grafik Tren Keuangan

Visualisasi chart tren pemasukan dan pengeluaran.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "ReportUI" as UI
control "ChartController" as Controller
entity "Database SQLite" as DB

title Sequence Diagram UC-13: Melihat Grafik Tren Keuangan

User -> UI: pilihTampilanGrafik()
activate UI
UI -> Controller: getChartData()
activate Controller

Controller -> DB: fetchTimeSeriesData()
activate DB

alt Data Seri Waktu Tersedia
    DB --> Controller: timeSeriesList
    Controller -> Controller: renderChartModel(timeSeriesList)
    Controller --> UI: displayLineOrBarChart(chartModel)
    UI --> User: tampilkanVisualisasiGrafik()
else Data Kosong
    DB --> Controller: emptyData
    deactivate DB
    Controller --> UI: displayFlatChart("Grafik Kosong")
    deactivate Controller
    UI --> User: tampilkanGrafikDatar()
    deactivate UI
end
@enduml
```

UC-14: Mengekspor Laporan Keuangan (Extended UC)

Pengunduhan berkas laporan ke format PDF/CSV.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "ReportUI" as UI
control "ExportController" as Controller
boundary "FileChooserUI" as FileChooser
entity "Disk Lokal" as Disk

title Sequence Diagram UC-14: Mengekspor Laporan Keuangan (Extended)

User -> UI: klikEksporLaporanPDFatauCSV()
activate UI
UI -> Controller: initExportProcess()
activate Controller

Controller -> FileChooser: openSaveDialog()
activate FileChooser
FileChooser --> User: tampilkanPilihFolderAndNamaFile()
User -> FileChooser: pilihLokasiAndSimpan(filePath)
FileChooser --> Controller: targetFilePath
deactivate FileChooser

Controller -> Disk: convertAndWriteToFile(targetFilePath, reportData)
activate Disk

alt Penulisan File Berhasil
    Disk --> Controller: writeSuccess
    Controller --> UI: pesanSukses("File Laporan Berhasil Diekspor")
    UI --> User: tampilkanNotifikasiEksporSukses()
else Penulisan File Gagal (Disk Penuh / Hak Akses Ditolak)
    Disk --> Controller: writeError
    deactivate Disk
    Controller --> UI: pesanError("Gagal Menulis File ke Disk")
    deactivate Controller
    UI --> User: tampilkanErrorPenyimpanan()
    deactivate UI
end
@enduml
```

UC-15: Melakukan Backup Basis Data Lokal

Penyalinan fisik berkas database .db.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "DatabaseUI" as UI
control "DatabaseController" as Controller
boundary "FileChooserUI" as FileChooser
entity "Database SQLite" as DB
entity "Disk Lokal" as Disk

title Sequence Diagram UC-15: Melakukan Backup Basis Data Lokal

User -> UI: klikBackupBasisData()
activate UI
UI -> Controller: initBackupProcess()
activate Controller

Controller -> FileChooser: openFolderChooser()
activate FileChooser
FileChooser --> User: tampilkanPilihFolderTujuan()
User -> FileChooser: pilihFolderTujuan(destinationPath)
FileChooser --> Controller: destinationPath
deactivate FileChooser

Controller -> DB: getActiveDatabaseFilePath()
activate DB
DB --> Controller: sourceDbPath
deactivate DB

Controller -> Disk: copyFile(sourceDbPath, destinationPath)
activate Disk

alt Penyalinan Berhasil
    Disk --> Controller: copySuccess
    Controller --> UI: pesanSukses("Backup Database Berhasil Disimpan")
    UI --> User: tampilkanNotifikasiBackupBerhasil()
else Penyalinan Gagal
    Disk --> Controller: copyFailed
    deactivate Disk
    Controller --> UI: pesanError("Gagal Melakukan Backup File")
    deactivate Controller
    UI --> User: tampilkanNotifikasiGagalBackup()
    deactivate UI
end
@enduml
```

UC-16: Melakukan Restore Basis Data Lokal

Pengembalian database dari file cadangan dengan penimpanan file lokal.

```
@startuml
autonumber
actor "Pengguna (User)" as User
boundary "DatabaseUI" as UI
control "DatabaseController" as Controller
boundary "FileChooserUI" as FileChooser
entity "Database SQLite" as DB

title Sequence Diagram UC-16: Melakukan Restore Basis Data Lokal

User -> UI: klikRestoreBasisData()
activate UI
UI --> User: tampilkanDialogKonfirmasiPenimpananData()

alt Pengguna Menyetujui Konfirmasi
    User -> UI: setujuPenimpananData("Ya")
    UI -> Controller: initRestoreProcess()
    activate Controller

    Controller -> FileChooser: openFileSelectionDialog()
    activate FileChooser
    FileChooser --> User: tampilkanPilihFileDB()
    User -> FileChooser: pilihFileBackup(backupFilePath)
    FileChooser --> Controller: backupFilePath
    deactivate FileChooser

    Controller -> Controller: validatesSQLiteFileSchema(backupFilePath)

    alt File Valid & Tidak Corrupted
        Controller -> DB: replaceActiveDatabaseFile(backupFilePath)
        activate DB
        Controller -> DB: restartDatabaseConnection()
        DB --> Controller: connectionRestored
        deactivate DB

        Controller --> UI: notifySuccessAndReloadUI("Restore Berhasil")
        UI --> User: muatUlangAntarmukaAndTampilkanSukses()

    else File Backup Rusak / Corrupted
        Controller --> UI: pesanError("File Database Tidak Valid")
        UI --> User: tampilkanErrorFileCorrupted()
    end

else Pengguna Menolak Konfirmasi
    User -> UI: setujuPenimpananData("Tidak")
    UI --> User: batalRestore()
    deactivate Controller
    deactivate UI
end
@enduml
```